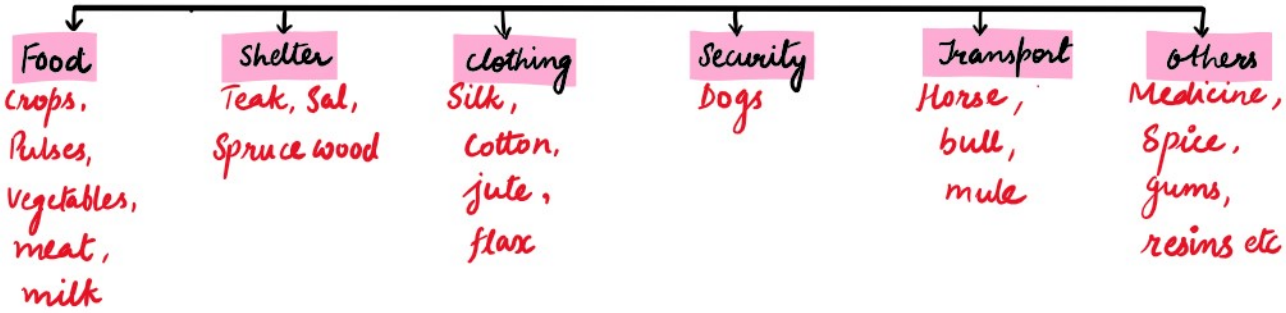
CLASSIFICATION

- We have identified 1·7 – 1·8 million species till date. ∴ To read and understand such large no. of organisms, classification is important.

Schemes of Classification

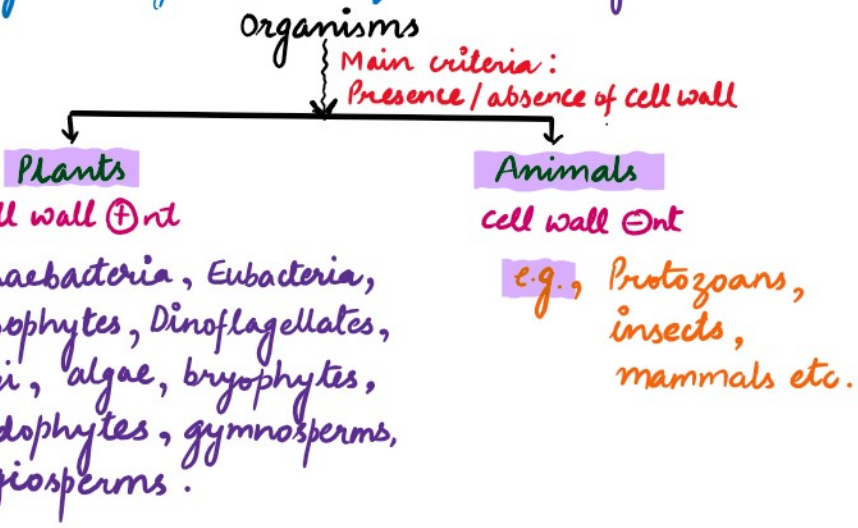
- Earliest classification by 'Early men' : It was a non-scientific classification based on 'uses' of organisms.



- First Scientific classification : By Aristotle



- Two Kingdom System classification : By Carolus Linnaeus



Angiosperms.

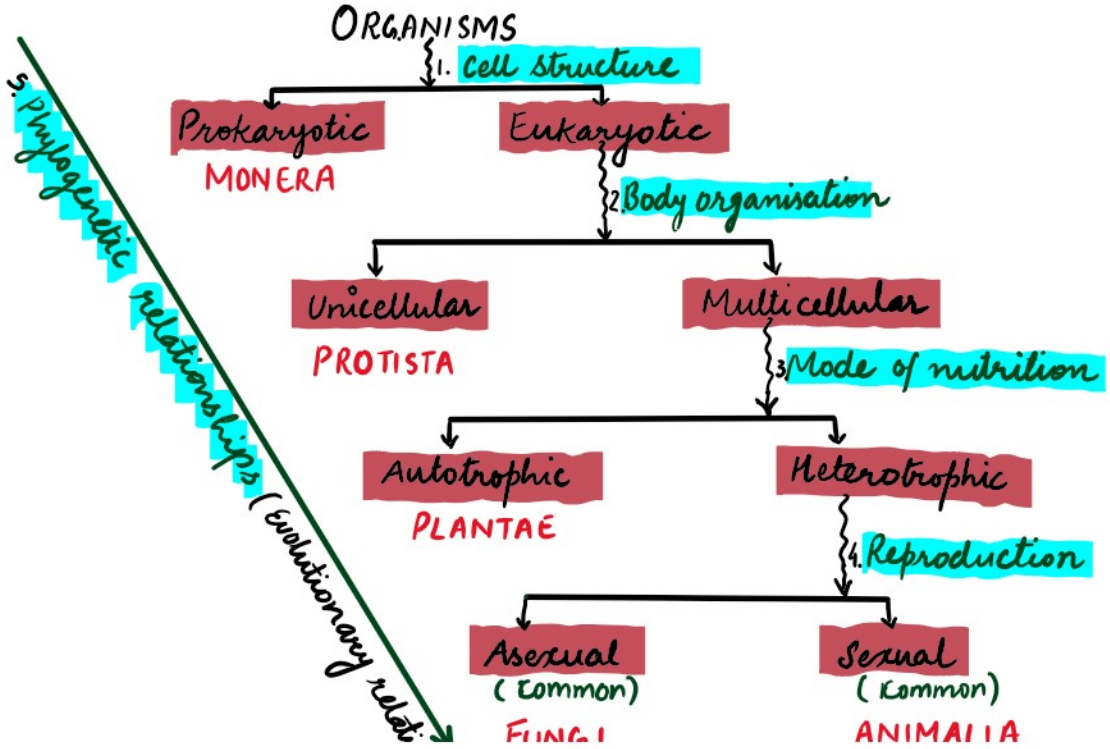
Drawbacks :

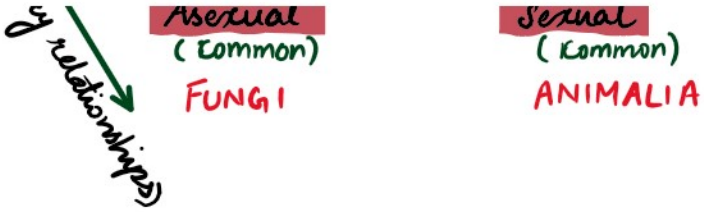
- This system did not distinguish between prokaryotic (bacteria, BGA), and eukaryotic (fungi, plants) organisms.
- It kept together unicellular (Chlorella) with multicellular plants.
- It kept together heterotrophic (fungi) and autotrophic (plants) organisms together.

NOTE: Some organisms like mycoplasma lacked cell wall but kept with bacteria in plantae.

- **3-Kingdom System** : Haeckel
  - Introduced Protista
- **4-Kingdom System** : Copeland
  - Introduced Monera
- **5-Kingdom System** : Given by R.H. Whittaker in 1969 and
  - he introduced Kingdom Fungi.

Main Criteria of classification :-





Characters	5 Kingdoms				
	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	←	Eukaryotic	→	
Cell wall	Polysaccharide + Amino acid	⊕ in some	Chitin	Cellulose	⊖ nt
Nuclear membrane	⊖ nt	←	⊕ nt	→	
Body Organisation	Cellular	Cellular	Multicellular/ loose tissue	Tissue/ organ	Tissue / organ / organ system
Mode of nutrition	Autotrophic/ heterotrophic	Autotrophic / Heterotrophic	Hetero-trophic	Auto-trophic	Hetero-trophic

NOTE

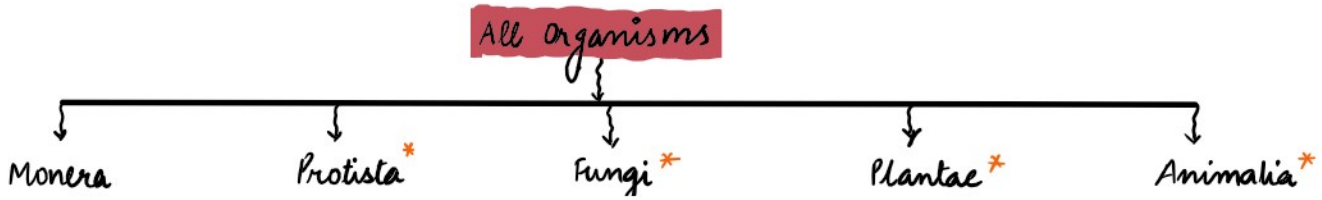
1. Chlamydomonas & Chlorella – earlier, were placed in Plants
   – Cell wall ⊕ nt
2. Amoeba & Paramoecium – earlier, were placed in animals
   – cell wall ⊖ nt

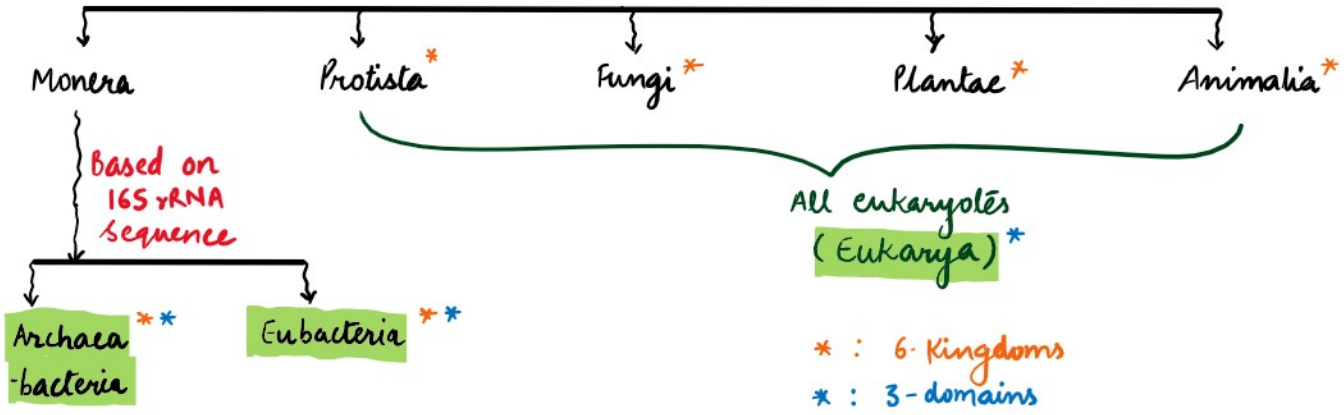
↓

Now kept in Protista (due to unicellular eukaryotic nature)

3. In classification system, Plant & Animal Kingdom were CONSTANT but organisms within them kept on changing due to change in criteria of classification.
   • such changes may occur in future too.

- 6-Kingdom or 3-domain system of classification : By Carl Woese




Monera
Protista*
Fungi*
Plantae*
Animalia*
Based on 16S rRNA sequence
All eukaryotes (Eukarya)*
Archaea -bacteria**
Eubacteria**
*: 6-Kingdoms
*: 3-domains

KINGDOM MONERA

- Bacteria are their sole members
- Bacteria are the most abundant & cosmopolitan (present everywhere) organisms.
- Hundreds of bacteria are present in handful of soil.
- Most of the bacteria are heterotrophic; only a few are autotrophic

Heterotrophic → Parasitic / Saprophytic (on dead & decaying matter)

Parasitic → Ectoparasite (Grow on organisms.) / Endoparasite (Grow inside organisms.)

Autotrophic → Photo-synthetic / Chemo-synthetic

- Bacteria are simple in structure and very complex in behaviour i.e., they show very complex metabolic diversity as they can do majority of reactions that a complex eukaryote can do.
- Prokaryotes can do some tasks that no eukaryote can do viz. N_2 - fixation.

Shapes of Bacteria

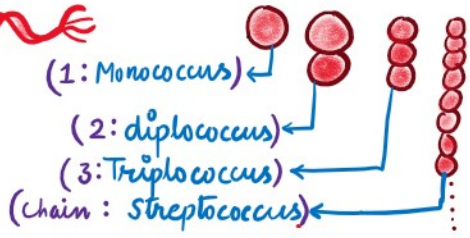
1. Bacillus (Rod-like)
2. Vibrio (comma-like)
3. Spirillum (Spring like)
4. Coccus (Round-shaped)



Endospore
(formed of Calcium-dipicolinic acid)



flagella







Sarcina: 8 cocci ar

→ (sheet : Staphyloccus)



Sarcina :. 8 cocci on the corner of a sphere

ARCHAEBACTERIA

- They are extremophiles due to their special cell wall structure.
  - ↳ have branched lipid
  - ↳ have ether linkage instead of ester linkage

extremophiles → Harsh condition / loving

- Halophiles → Salt loving
- Thermo-acidophiles
  - Thermo → ↑ Temperature :. found in thermal springs
  - acido → ↑ acidity ↓ pH
- Methanogens → CH_4 generating
  - Present in gut of ruminants like cow and buffalo & produce CH_4 from dung of these animals :. BIOGAS ↑

EUBACTERIA :

True

:. They are bacteria having rigid cell wall and flagella (if motile).

NUTRITION

Autotrophic

They can synthesise their own food (only some bacteria)

1. Photosynthetic autotrophs

Heterotrophic

Can't make their own food (most bacteria)

- Majority are decomposers i.e.,

own food (only some bacteria)

1. **Photosynthetic autotrophs**

Like cyanobacteria or BGA

↓

Means blue

- BGA has pigment 'chl-a' similar the higher plants.
- BGA can be - unicellular
  - colonial (forming colony)
  - Filamentous → (surrounded by Mucilage)

Generally made from collagen fibres & do provide protection to BGA-colonies.

- BGA can also be found in aquatic (both fresh and marine water) & Terrestrial habitat.
- They form bloom in polluted water. Polluted water has more nutrients to promote algal growth.
- Some cyanobacterias like Anabaena, Nostoc can fix nitrogen in specialised cells called heterocysts.

(most bacteria)

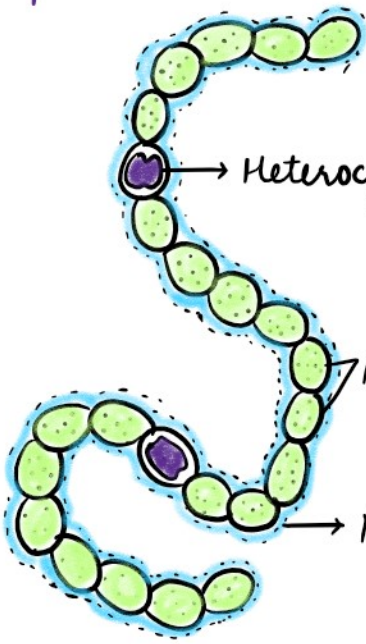
- Majority are decomposers i.e., grow on dead & decaying matter.
- They have significant impact on human affairs.

1) Making curd from milk e.g., Lactobacillus

2) Making antibiotics like Streptomycin & neomycin
   - by Streptomyces griseus
   - by Streptomyces fradiae

3) Fixing N_2 in legume roots i.e., in roots of leguminous plants like pea. e.g., Rhizobium

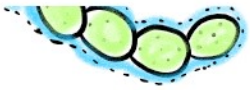
4) Some are pathogenic & cause diseases in animals & plants.
   - Cholera : Vibrio cholarae
   - Typhoid : Salmonella typhi
   - Citrus canker : Xanthomonas citri
   - Tetanus : Clostridium tetani



→ Heterocyst cell : Its special thickening do not let O_2 to come inside these cells ∵ Main enzyme for N_2-fixation i.e., nitrogenase gets inhibited by O_2

→ Normal vegetative cells

→ Mucilagenous sheath : Generally polysaccharide and glycoproteinaceous; it has pigments ∴ Provide colour to BGA; it is sticky & protect BGA from dessi-



it has pigments ∴ Provide colour to BGA; it is sticky & protect BGA from dessication (drying).

NOTE : Mucilagenous & gelatenous coats are different in nature but both can be present on BGA.

2. **Chemosynthetic autotrophs** :

- Oxidise various inorganic chemicals to release energy for the formation of ATP.
- Nitrite, Nitrate, Ammonia ⟶ Energy + Nitrogen
  Phosphate ⟶ Energy + Phosphorous
  Ferrate ⟶ Energy + Iron
  Sulphate ⟶ Energy + Sulphur
  } They help in nutrient recycling

REPRODUCTION IN BACTERIA

1. Mainly by Fission : Here, a bacteria divides into 2-daughter bacterium.
   - Initially, the genetic material (DNA) gets replicated.
   - Then, the cell is divided into 2-parts.

2. Spore formation : during unfavourable conditions.

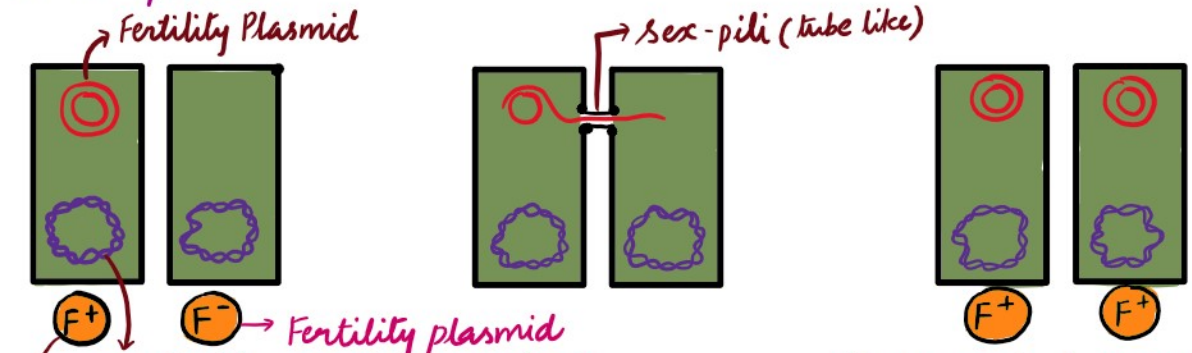


→ Endospore (resistant ∴ help to cope up with unfavourable conditions)

3. Sexual reproduction : This is primitive type of DNA transfer from one bacterium to another.
   Its called conjugation in bacteria which is not true sexual reproduction as no gamete formation & fusion takes place.

→ Fertility Plasmid

→ sex-pili (tube like)



∵ Fertility plasmid is ⊕nt ∴ called donor

At the end, both cells will become F^+ as single strand of DNA can make its complimentary strand.

MYCOPLASMA

- Smallest living cells : length = 0.3 μm
- PPLO is also a type of mycoplasma & have size = 0.1 μm

∵ They lack a rigid cell wall, they can change their shape & thus called 'Pleuomorphic'.

↓

They can change their shape just like a joker changes his expressions

∴ Mycoplasma are called as Joker of Plant Kingdom.

∵ In 2-Kingdom system of classification, mycoplasma were considered plants.

- They can live in aerobic conditions but can live without oxygen too ∴ they are facultative anaerobic (i.e., they switch to anaerobic conditions).
- Many mycoplasma can cause disease in plants and animals

  - Little leaf disease in brinjal
  - Witches Broom
  - Phylloidy (flowers become leaf-like)

  - Pneumonia like disease by PPLO
  - Arthritis (joint related disease)

KINGDOM PROTISTA

- contains unicellular, eukaryotic organisms.
- Members of Protista are primarily aquatic (marine + fresh)
- Boundaries of this Kingdom are not well defined.
  e.g. Chlorella, Chlamydomonas → According to a botanist, they are plants as they are photosynthetic and have cellulosic cell wall.
  ↳ According to a Protist Scientist they are protists due to their unicellular, eukaryotic nature

- They reproduce → Asexually: by fission (simple reproduction)
  → Sexually: by cell fusion (Zygote formation)

Monera	Protista	Fungi	Plantae	Animalia
Unicellular	Unicellular	multicellular		
Prokaryotes	Eukaryotes	Eukaryotes		

∴ Protista forms a link between monera & all other eukaryotic kingdoms.

Chrysophytes Plant-like	Dinoflagellates 2 flagella	Euglenoids connecting link between plants & animals	Slime Moulds fungus like ∴ Saprophytic	Protozoans animal-like
Photosynthesis↑ (∵ diatoms are chief Producers of ocean)	Mostly photo-synthetic	Mixotrophic • Photosynthetic like plants in presence of light Chl-a ⊕nt • Predator like animals: in absence of light	↓ Grow on dead and decaying twigs	↓ Heterotrophic and Parasitic
Cell wall present; embedded with	Stiff cellu-losic CW	Instead of CW, they have a	Usually CW is not ⊕nt	CW ⊖nt

cell wall present, embedded with Silica	Suff cellu- losic CW ⊕nt	Instead of CW, they have a protein rich layer called Pellicle ∴ Flexi- -bility ↑	...is not ⊕nt	CW ⊕nt
⊕nt in fresh & marine water	In marine water	In fresh and stagnant water	Numerous	Numerous

Chrysophytes :

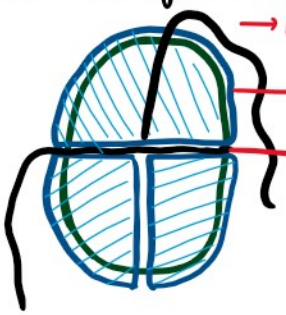
- They are consisted of : Diatoms and Desmids (Golden algae)
  - Diatoms: 2 Parts
- Silica deposition on cell wall that makes it indestructable
  - → Due to deposition of this indistructible material on base of ocean for billions of years due to death of diatoms: 'diatomaceous earth' is formed.



Due to its gritty (rough) nature, it is :
- Used in polishing
- Used in filtration of oils & syrups

- Being unicellular and photosynthetic ; these chrysophytes are called 'Phytoplanktons'.
  - → Float passively ∴ go along with the flow of ocean due to small size.

Dinoflagellates :



- They can be of different colours due to different pigments:
  - Yellow : due to carotenoid
  - Green : Chlorophyll
  - Brown : Xanthophyll (fucoxanthin)
  - Red : Erythrine

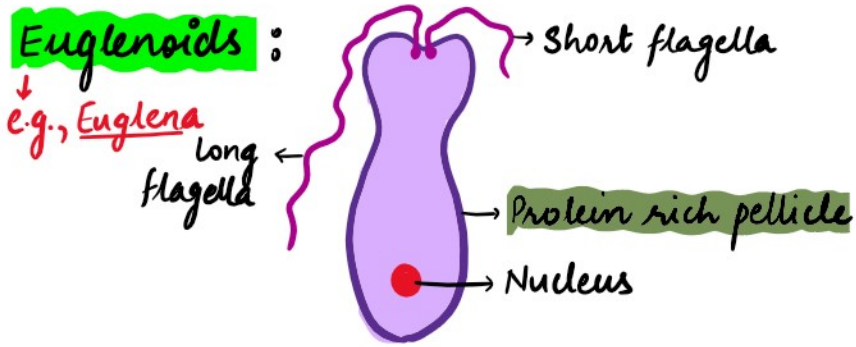
Gonyaulax cause 'RED TIDES'

This red dinoflagellate divides at a rapid rate ∴ Tides appear red

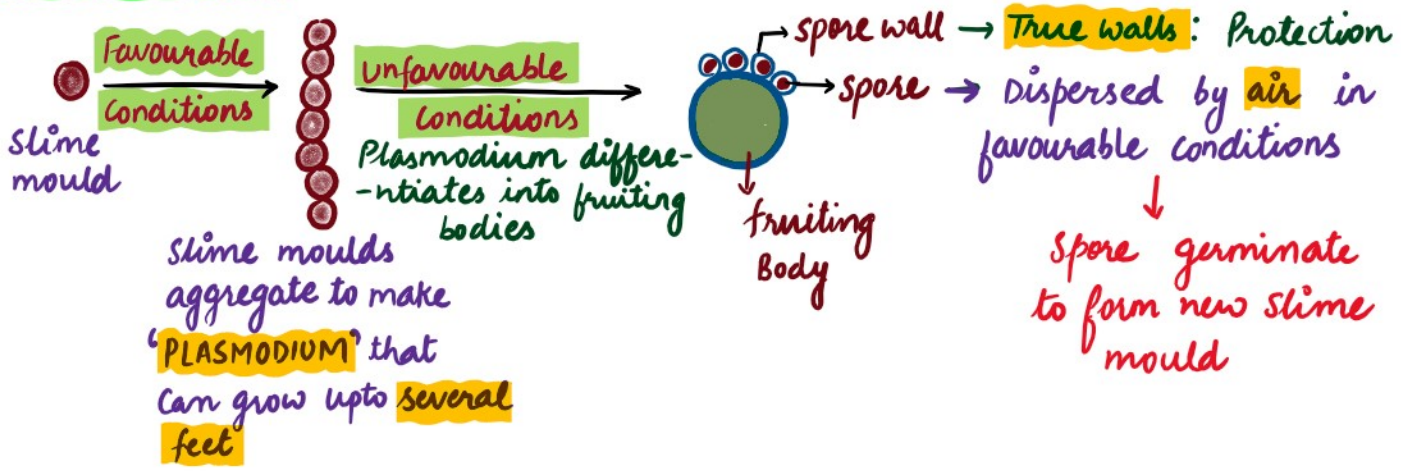
It secretes some toxins called saxi-toxins that can kill other marine animals nearly.

Euglenoids :

e.g., Euglena

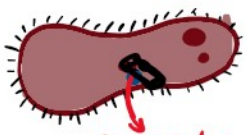


Slime Moulds :



Protozoans :

Amoeboid	Ciliated	Flagellated	Sporozoans
move by psedo-podia (false feet)	Thousands of cilia present that beat in coordinated way	Flagella present	Spore-like infectious stage is present
Can be found in	They are aquatic	They can be	Usually parasitic

Can be found in fresh water, marine water or moist soil	They are aquatic & they move actively because of cilia.	They can be free living or parasitic	Usually parasitic like Plasmodium (malarial parasite).
Marine forms have silica shells on their surface e.g., Entamoeba histolytica (cause amoebic dysentery)	Binucleated Paramoecium  Gullet for entry of food & water	e.g., Trypanosoma cause African Sleeping Sickness	Plasmodium can be of different types.

KINGDOM: FUNGI

- First antibiotic 'Penicillin' is isolated from Penicillium notatum which is a fungus.
- Kingdom fungi contain eukaryotic organisms that possess cell wall of chitin (a polymer of N-acetyl-glucosamine and polysaccharides)
- They are mostly multicellular and heterotrophic in nature.

Fungi are generally filamentous except a few like yeast (Saccharomyces) which is unicellular

- Called Baker's Yeast when used to make bread in bakery.
- Called Brewer's Yeast when used to make alcohol.

They can be:

Saprophytic
- On dead and decaying matter

or

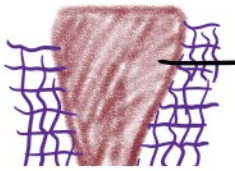
Parasitic
- They live in or on other organism

Endoparasites (In)
Ectoparasites (On)

- White spots on mustard due to: Albugo candida
- Black rust of wheat due to Puccinia graminis
- Fungal rashes and infections in animals

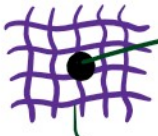
- Mushrooms (edible) and poisonous Toadstools (like Amantia) are also fungi.
- Orange rot and bread mould are also fungi.
- Fungi is cosmopolitan ∴ can be found in soil, air, water etc. but most favourable conditions for their growth is warm & humid conditions.
- Fungi also make associations with other organisms for food.

Mycorrhiza
Fungi + Root



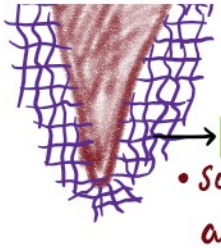
Root of Pinus
↓
will give food to fungi

Lichen: Association between Fungi and algae



Algae (Phycobiont)
- Give food to fungi

Fungi (Mycobiont)
- ↑ surface area for H2O



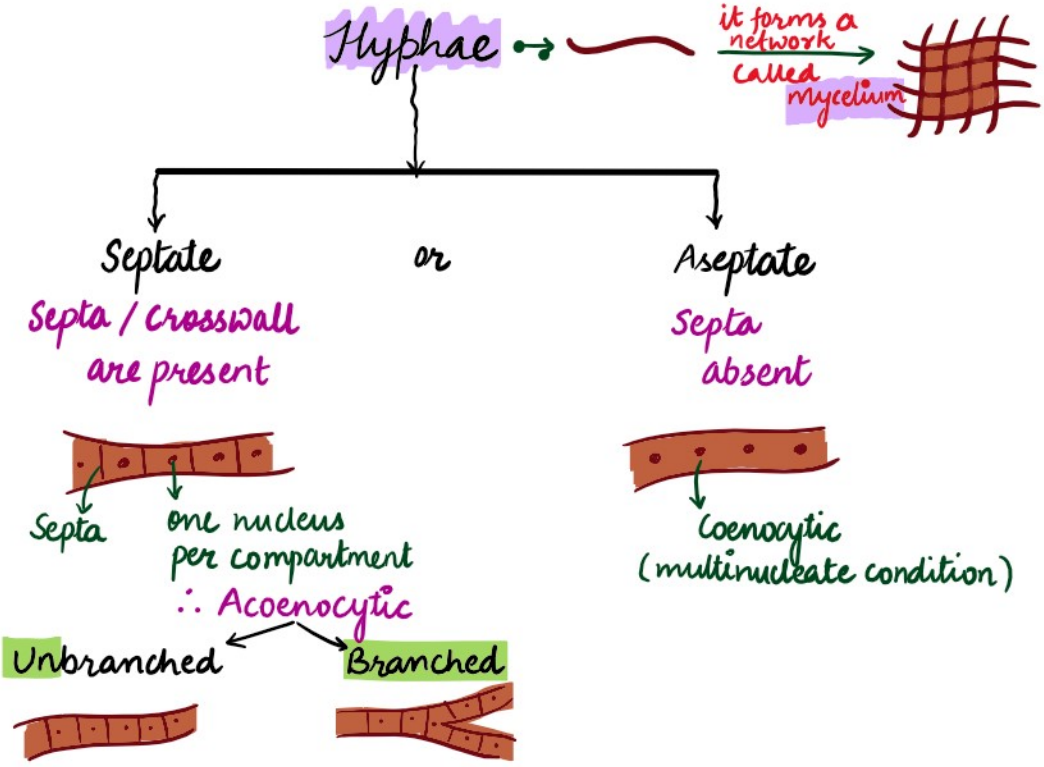
Root of Plants
↓
Will give food to fungi

→ Fungi
- Surface area ↑ for water and mineral absorption
- Provide protection / covering to root against pathogens

→ Fungi (Mycobiont)
- ↑ surface area for H_2O & mineral absorption
- Provide shelter to algae

NOTE: Lichen are sensitive to SO_2 ∴ they can't grow in polluted areas.

- Fungi are filamentous and they are formed of thin, slender, long, thread-like structures called hyphae.



NOTE: Coenocytic hyphae is found in phycomycetes and septate, branched hyphae is found in rest all i.e., ascomycetes, basidiomycetes & deuteromycetes.

Reproduction in Fungi

(i) Vegetative reproduction :- By fission,
- budding,
- fragmentation

- budding,
- fragmentation

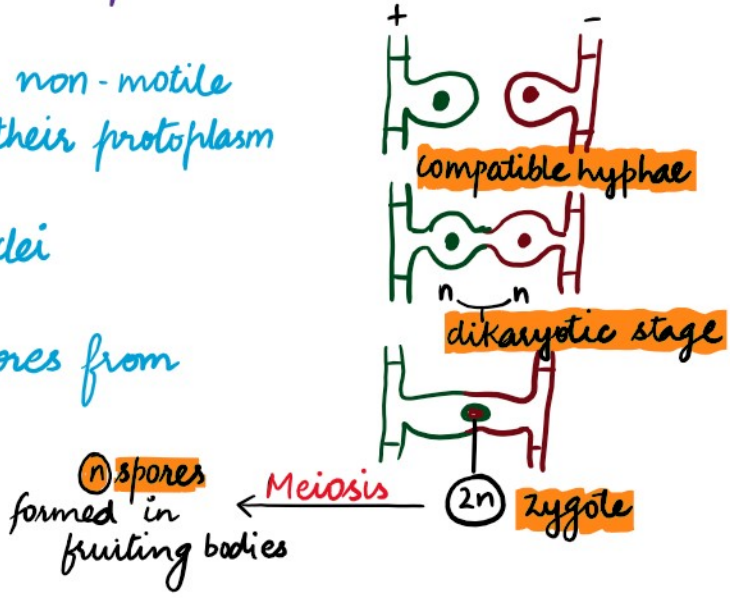
(ii) Asexual reproduction: - By asexual spores like - conidia,
- Zoospore,
- aplanospore,
- sporangiospore

(iii) Sexual reproduction: By sexual spore like - ascospores,
- basidiospores
- Oospores
- Zygospores

↓
• Gamete fusion & zygote formation takes place

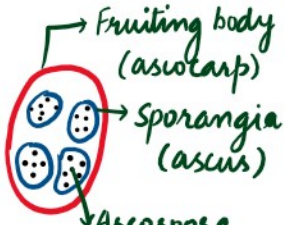
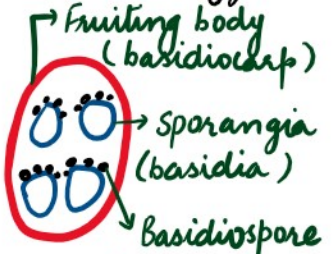
• There are three-steps in sexual reproduction:

1. **Plasmogamy**: Two motile or non-motile gametes come together & their protoplasm fuses.
2. **Karyogamy**: Fusion of nuclei
3. **Meiosis**: formation of (n) spores from (2n) zygote



* Dikaryophase is more prolonged in ascomycetes & basidiomycetes.

Phycomycetes	Ascomycetes	Basidiomycetes	Deuteromycetes
Obligate Parasites on plants	Sac-fungi e.g., Saccharomyces • They are coprophiles i.e., dung loving ∴ decompose dung	e.g., common Mushroom ↓ Reproduce mainly by fragmentation. ∴ asexual spores are usually not found	↓ Decomposers (organic to inorganic) ∴ do mineral cycling
Coenocytic & aseptate hyphae	Septate & branched hyphae	Septate & branched hyphae	Septate & branched hyphae
Aquatic: found in	They are saprophytic,	Found in soil, log	Found on dead

Aquatic : found in warm & humid conditions like decaying wood	They are saprophytic, Parasitic & decomposers Antibiotic : Penicillium Edible : Morels & Truffles	Found in soil, log of wood, moist & damp conditions	Found on dead organic matter
Reproduction: • Vegetative : fragmentation (very common) • Asexual : By spores Zoospores (motile); Aplanospore (non-motile) ↳ endogenous development → sporangia → spores Sexual : Iso or anisogametes fuse & zygospore formed e.g., Rhizopus (bread mould) Mucor, Albuga	Asexual : Conidia  developed exogenously Ascocarps are cup-shaped Sexual : Ascospores  Fruiting body (ascocarp) Sporangia (ascus) Ascospore (endogenous) e.g., Neurospora, Aspergillus, Cleviceps (make LSD & alkaloid)	• Sexual reproduction is present but sex organs are absent ↓ ∴ 2 somatic or vegetative cells will fuse to make zygote.  Fruiting body (basidiocarp) Sporangia (basidia) Basidiospore (exogenous) e.g., Agaricus (Mushroom) Ustilago (Smut) Puccinia (rust)	• Called imperfect fungi (as sexual reproduction is absent) ↓ Asexual reproduction done only by conidia e.g., Alternaria, Colletotrichum, Trichoderma

NOTE : Neurospora is called as Drosophila of Plant Kingdom, as it is used widely in experiments of genetics & biochemistry.

NOTE : Both spores starting from 'A' i.e., Aplanospore & Ascospore are endogenous.

KINGDOM PLANTAE

- Kingdom Plantae includes all eukaryotic chlorophyll-containing organisms commonly called plants.
  - Photo-synthetic
  - ∴ Autotrophic
- A few members are partially heterotrophic
  - Insectivorous Plants (e.g., Bladderwort, Venus fly trapa)
  - Parasites (e.g., Cuscuta)
- cellulosic cell wall.
- Life cycle has 2-different phases
  - (2n) sporophytic
  - (n) gametophitic

KINGDOM ANIMALIA

- Characterised by heterotrophic eukaryotic organisms.
- Includes multicellular organisms.
- Cells lack cell walls.
- They are directly or indirectly depend on plants for food.
- They digest their food in an internal cavity and store food reserves as glycogen or fat.
- Their mode of nutrition is holozoic - by ingestion of food.
- They follow a definite growth pattern & grow into adults that have a definite shape and size.
- Higher forms show elaborate sensory and neuromotor mechanism.
- Most of them are capable of locomotion.
- Sexual reproduction : By copulation of male & female followed by embryological development.

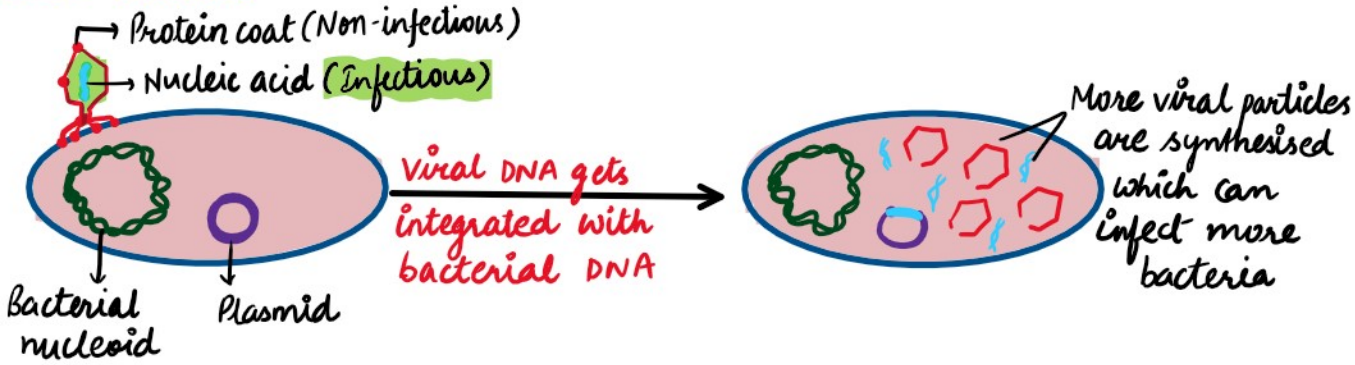
VIRUS, VIROID AND PRIONS

- They all do not have a cellular organisation
  ∴ are non-living
- They are not placed under any kingdom by R.H.Whittaker

VIRUS

- They are obligate (compulsory) parasites, as they behave like inert crystalline particles outside host cell.

- But, when they infect host cell, they hijacks or take over the machinary of host cell and they instruct host cell to make more Viruses.



- D.J. Ivanowsky (1892) : Discovered virus; identified that mosaic disease of tobacco is due to an agent which is smaller than bacteria (i.e., it can cross bacterial proof filters)
- M.W. Beijerinck (1898) : Called virus a "Living Infectious Agent" or "Contagium vivum fluidum".

- W.M. Stanley (1935) : He crystallised virus and said virus crystals are made largely of proteins.

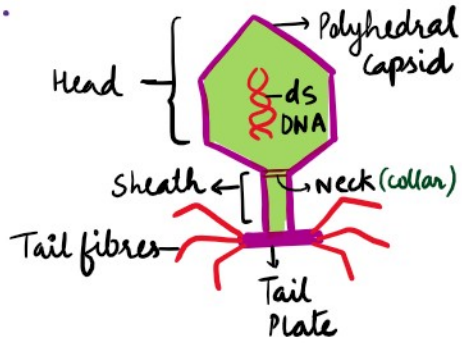
- Virus is a nucleo- protein particle
  - nucleo → Nucleic acid
  - protein → Protein
    - It covers the genetic material & is called 'CAPSID' which is
  - Any virus can contain

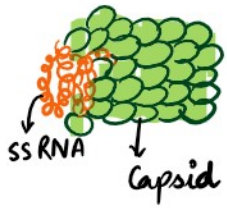
acid

- Any virus can contain either DNA or RNA, but never both.
- Every virus contains specific genetic material :
  - → Bacteriophage : ds DNA
  - → Plant viruses : ss RNA
  - → Animal viruses : ds DNA, ss RNA, ds RNA.

↳ It covers the genetic material & is called 'CAPSID', which is made of many small sub-units called 'CAPSOMERES'

- Can be arranged 'Polyhedrally' e.g., in Bacteriophage
- Can be arranged 'helically' e.g., in TMV (Tobacco Mosaic Virus)





Symptoms of viral diseases in Plants :

- stunted growth ; dwarfness ; mosaic formation (spots) ; leaf rolling ; leaf curling ; vein clearing ; chlorosis (loss of chlorophyll) and necrosis (tissue death).

- Common diseases caused by Viruses in animals :
  Herpes, AIDS, Influenza, Mumps, small pox, Measles, Rubella, Polio.

VIROIDS

- They are small RNA molecules of low molecular weight without capsid.
- Discovered by T.O. Diener in 1971.
- Viroids are smaller than virus as they lack protein coat.
- It cause Potato Spindle Tuber Disease (PSTD) in potato due to which potato become spindle shaped (thin) ∴ yield ↓.

PRIONS

- Proteinaceous infectious particles having size similar to virus.
- Abnormally folded protein can accumulate in our brain ∴ cause several diseases
- Mad cow disease or Bovine spongiform encephalopathy (BSE) in cattles.
- Cruetzfeldt (Cr) Jacob disease in man.

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